

# (MATH000) Mathematical Language

*Notation and Conventions*

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## Part I. General Mathematical Language

### 1 General Conventions

These conventions apply throughout. Widely used standard alternatives may be listed for the same concept in preferred order; the first is used throughout.

| Item                       | Convention                                                                                                                                                                                                       |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scalars                    | Ordinary mathematical italic letters, for example $a, b, c, \lambda, t$ .                                                                                                                                        |
| Vectors                    | Bold lowercase letters, for example $\mathbf{u}, \mathbf{v}, \mathbf{x}$ . Unless otherwise stated, vectors are column vectors.                                                                                  |
| Matrices                   | Bold uppercase letters, for example $\mathbf{A}, \mathbf{B}, \mathbf{M}$ .                                                                                                                                       |
| Dimension subscripts       | Prefer dimension subscripts when the dimension is readily available and useful; omit them when it is clear, immaterial, or not readily available rather than inferring a dimension solely to supply a subscript. |
| Greek vectors and matrices | Bold Greek letters when a Greek symbol denotes a vector or matrix, for example $\boldsymbol{\xi}$ and $\boldsymbol{\Phi}$ .                                                                                      |
| Number systems             | Blackboard bold, for example $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ .                                                                                                                      |
| Sets                       | Ordinary uppercase letters such as $A, B, X, Y$ , unless a field-specific convention requires otherwise.                                                                                                         |
| Families and collections   | Calligraphic letters when useful, for example $\mathcal{F}, \mathcal{T}, \mathcal{B}$ .                                                                                                                          |
| Named operators            | Upright Roman type, for example $\det, \ker, \operatorname{im}, \operatorname{rk}$ , and $\operatorname{tr}$ .                                                                                                   |
| Large operators            | Display size in displayed mathematics and notation tables; sums and products place indices above and below, limits place subscripts below, and integral bounds retain standard right-side placement.             |
| Greek variants             | Distinct glyphs represent distinct symbols; in particular, $\varepsilon$ and $\epsilon$ are not interchangeable.                                                                                                 |

| Notation                    | Meaning                                                               |
|-----------------------------|-----------------------------------------------------------------------|
| $(a_1, \dots, a_n)$         | Ordered tuple; parentheses are also the default matrix delimiters.    |
| $(a, b)$                    | Open interval.                                                        |
| $[a, b]$                    | Closed interval.                                                      |
| $[a, b)$                    | Left-closed, right-open interval.                                     |
| $(a, b]$                    | Left-open, right-closed interval.                                     |
| $\{a_1, \dots, a_n\}$       | Set delimiters.                                                       |
| $\{x \in A \mid P(x)\}$     | Set-builder notation.                                                 |
| $\langle u, v \rangle$      | Pairing or inner product, according to context.                       |
| $ x $                       | Absolute value, modulus, cardinality, or order, according to context. |
| $\ x\ $                     | Norm.                                                                 |
| s.t.                        | Abbreviation for “such that” when naturally required.                 |
| $a = b$                     | Equality.                                                             |
| $a := b$                    | $a$ is defined to be $b$ .                                            |
| $a \equiv b$                | Identity, equivalence, or congruence, according to context.           |
| $a \cong b$                 | Isomorphism or congruence, according to context.                      |
| $a \approx b$               | Approximation.                                                        |
| $f: X \rightarrow Y$        | Function or map from $X$ to $Y$ .                                     |
| $x \mapsto f(x)$            | Elementwise rule of a map.                                            |
| $g \circ f$                 | Composition: first $f$ , then $g$ .                                   |
| $f _A, f \upharpoonright_A$ | Restriction of $f$ to $A$ .                                           |

| Notation      | Meaning               |
|---------------|-----------------------|
| $\text{id}_X$ | Identity map on $X$ . |

## 2 Theorem-like Environments

| Item           | Convention                                                                   |
|----------------|------------------------------------------------------------------------------|
| Definition     | Precise meaning of a term, object, property, or construction.                |
| Theorem        | Principal mathematical result.                                               |
| Proposition    | Useful result of more limited scope.                                         |
| Lemma          | Supporting result, usually for a later proof.                                |
| Corollary      | Result following readily from a preceding result.                            |
| Conjecture     | Statement proposed to be true but not proved in full generality.             |
| Hypothesis     | Named assumption or proposed statement.                                      |
| Formula        | Distinguished computational identity or closed form.                         |
| Algorithm      | Explicit step-by-step procedure.                                             |
| Fact           | Verified statement not requiring theorem status.                             |
| Example        | Illustration of a definition, construction, phenomenon, or result.           |
| Counterexample | Example disproving a general assertion or showing a hypothesis is necessary. |
| Remark         | Supplementary context, warning, or qualification.                            |
| Problem        | Mathematical task intended to be solved.                                     |
| Open Problem   | Problem not known to be solved in the stated form.                           |
| Idea           | Decisive conceptual insight.                                                 |
| Strategy       | Broad problem-solving plan.                                                  |
| Tactic         | Local step within a larger strategy.                                         |

Proof labels are written as *Proof* and solution labels as *Solution*, each followed by a period.

| Notation | Meaning                                                                 |
|----------|-------------------------------------------------------------------------|
| □        | Ending for ordinary proofs and solutions.                               |
| ■        | Ending for substantial, difficult, or particularly important proofs.    |
| Q.E.D.   | Ending for exceptionally difficult, significant, or paper-level proofs. |

**Enumeration and Subdivision.** Enumeration labels are set in upright Roman type. Their form is determined by structural role rather than nesting depth.

| Item                  | Convention                                                                                                                                                            |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1, 1.2, 1.3, ...    | Independent child problems or standalone units under a common parent; each may be referenced separately.                                                              |
| (1), (2), (3), ...    | Ordinary enumerated items, such as choices, cases, examples, or procedural steps.                                                                                     |
| (a), (b), (c), ...    | Connected subparts of a single problem sharing a common setup.                                                                                                        |
| (i), (ii), (iii), ... | Formal mathematical clauses, such as hypotheses, conditions, properties, or equivalent statements within definitions, theorems, propositions, and related statements. |

The distinction between 1.1 and (a) is independence: 1.1 denotes a standalone child unit, whereas (a) denotes a connected subpart of one problem. Likewise, (1) is used for ordinary

enumeration, while (i) is reserved for formal mathematical clauses. Nested enumeration preserves these roles; for example, a subpart (a) may contain formal clauses (i), (ii), . . . .

| Item                  | Convention                                                                                  |
|-----------------------|---------------------------------------------------------------------------------------------|
| Displayed mathematics | Remains part of the surrounding sentence and receives punctuation when grammar requires it. |
| Equation numbering    | Number equations that are referenced later or benefit from a stable label.                  |
| Logical prose         | Use words such as “and” and “or” unless symbolic logic improves clarity.                    |
| Abbreviations         | Use established abbreviations only when standard and unambiguous.                           |

## Part II. Notation by Mathematical Area

### 3 Set Theory and Logic

Uppercase letters usually denote sets; lowercase letters usually denote their elements.

| Notation                               | Meaning                                                       |
|----------------------------------------|---------------------------------------------------------------|
| $\mathbb{N} = \{1, 2, 3, \dots\}$      | Positive integers.                                            |
| $\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$ | Nonnegative integers.                                         |
| $\mathbb{Z}$                           | Integers.                                                     |
| $\mathbb{Q}$                           | Rational numbers.                                             |
| $\mathbb{R}$                           | Real numbers.                                                 |
| $\mathbb{C}$                           | Complex numbers.                                              |
| $\emptyset, \varnothing$               | Empty set.                                                    |
| $x \in A$                              | $x$ is an element of $A$ .                                    |
| $x \notin A$                           | $x$ is not an element of $A$ .                                |
| $A \subseteq B$                        | $A$ is a subset of $B$ ; equality is allowed.                 |
| $A \subsetneq B$                       | $A$ is a proper subset of $B$ .                               |
| $A \supseteq B$                        | $A$ is a superset of $B$ ; equality is allowed.               |
| $A \supsetneq B$                       | $A$ is a proper superset of $B$ .                             |
| $A \cup B$                             | Union.                                                        |
| $A \cap B$                             | Intersection.                                                 |
| $\bigcup_{i \in I} A_i$                | Union of an indexed family of sets.                           |
| $\bigcap_{i \in I} A_i$                | Intersection of an indexed family of sets.                    |
| $A \setminus B, A - B$                 | Set difference.                                               |
| $A \triangle B, A \Delta B$            | Symmetric difference.                                         |
| $A^c$                                  | Complement of $A$ in the ambient set.                         |
| $\mathcal{P}(A), 2^A$                  | Power set of $A$ .                                            |
| $A \times B$                           | Cartesian product.                                            |
| $A \sqcup B, A \dot{\cup} B$           | Disjoint union.                                               |
| $A \cap B = \emptyset$                 | $A$ and $B$ are disjoint.                                     |
| $ A , \#A$                             | Cardinality.                                                  |
| $\{x \in A \mid P(x)\}$                | Elements $x \in A$ satisfying $P(x)$ .                        |
| $x \sim y$                             | Equivalent elements under an understood equivalence relation. |
| $A/\sim$                               | Set of equivalence classes.                                   |

| Notation                    | Meaning                                  |
|-----------------------------|------------------------------------------|
| $\forall x \in A$           | For every $x \in A$ .                    |
| $\exists x \in A$           | There exists $x \in A$ .                 |
| $\exists! x \in A$          | There exists a unique $x \in A$ .        |
| $\nexists x \in A$          | There does not exist $x \in A$ .         |
| $P \Rightarrow Q$           | $P$ implies $Q$ .                        |
| $P \Leftarrow Q$            | $Q$ implies $P$ .                        |
| $P \nRightarrow Q$          | $P$ does not imply $Q$ .                 |
| $P \Leftrightarrow Q$       | $P$ if and only if $Q$ .                 |
| $P \wedge Q$                | Logical conjunction.                     |
| $P \vee Q$                  | Logical disjunction.                     |
| $\neg P$                    | Logical negation.                        |
| $\text{dom } f$             | Domain of $f$ .                          |
| $\text{codom } f$           | Codomain of $f$ .                        |
| $f(A)$                      | Image of $A$ under $f$ .                 |
| $f^{-1}(B)$                 | Preimage of $B$ under $f$ .              |
| $\text{im } f$              | Image of the map $f$ .                   |
| $f: X \hookrightarrow Y$    | Injective map.                           |
| $f: X \twoheadrightarrow Y$ | Surjective map.                          |
| $f: X \xrightarrow{\sim} Y$ | Isomorphism; for sets, a bijection.      |
| $\mathbb{1}_A(x)$           | Indicator function of $A$ .              |
| $\aleph_0$                  | Cardinality of a countably infinite set. |
| $\mathfrak{c}$              | Cardinality of the continuum.            |

## 4 Calculus and Analysis

The differential  $\mathrm{d}$  is upright;  $\varepsilon$  is the default small positive quantity in analysis. Function spaces take their domains in parentheses, for example  $C([a, b])$ ,  $\mathcal{R}([a, b])$ , and  $L^p(\Omega)$ .

| Notation                                           | Meaning                                    |
|----------------------------------------------------|--------------------------------------------|
| $\lim_{x \rightarrow a} f(x)$                      | Limit as $x$ approaches $a$ .              |
| $\lim_{x \rightarrow a^-} f(x)$                    | Left-hand limit as $x$ approaches $a$ .    |
| $\lim_{x \rightarrow a^+} f(x)$                    | Right-hand limit as $x$ approaches $a$ .   |
| $\min A$                                           | Minimum of $A$ .                           |
| $\max A$                                           | Maximum of $A$ .                           |
| $\inf A$                                           | Infimum of $A$ .                           |
| $\sup A$                                           | Supremum of $A$ .                          |
| $f'(a)$                                            | First derivative of $f$ at $a$ .           |
| $f^{(n)}(a)$                                       | $n$ th derivative of $f$ at $a$ .          |
| $\frac{\mathrm{d}y}{\mathrm{d}x}$                  | Derivative of $y$ with respect to $x$ .    |
| $\frac{\partial f}{\partial x_i}$ , $\partial_i f$ | Partial derivative with respect to $x_i$ . |
| $\int_a^b f(x) \, \mathrm{d}x$                     | Definite integral over $[a, b]$ .          |
| $\iint_D f \, \mathrm{d}A$                         | Double integral over $D$ .                 |
| $\iiint_\Omega f \, \mathrm{d}V$                   | Triple integral over $\Omega$ .            |

| Notation                                                       | Meaning                                    |
|----------------------------------------------------------------|--------------------------------------------|
| $\oint_C f(z) \, dz$                                           | Contour integral over a closed curve $C$ . |
| $\sum_{k=1}^n a_k$                                             | Finite sum.                                |
| $\sum_{k=1}^{\infty} a_k$                                      | Infinite series.                           |
| $\prod_{k=1}^n a_k$                                            | Finite product.                            |
| $e^x, \exp x$                                                  | Exponential function.                      |
| $\ln x, \log_e x$                                              | Logarithm to base $e$ .                    |
| $\lg x, \log_{10} x$                                           | Logarithm to base 10.                      |
| $\text{lb } x, \log_2 x$                                       | Logarithm to base 2.                       |
| $\log_a x$                                                     | Logarithm to base $a$ .                    |
| $\sin x$                                                       | Sine.                                      |
| $\cos x$                                                       | Cosine.                                    |
| $\tan x$                                                       | Tangent.                                   |
| $\csc x$                                                       | Cosecant.                                  |
| $\sec x$                                                       | Secant.                                    |
| $\cot x$                                                       | Cotangent.                                 |
| $\arcsin x, \sin^{-1} x$                                       | Principal inverse sine.                    |
| $\arccos x, \cos^{-1} x$                                       | Principal inverse cosine.                  |
| $\arctan x, \tan^{-1} x$                                       | Principal inverse tangent.                 |
| $\text{arccsc } x, \csc^{-1} x$                                | Principal inverse cosecant.                |
| $\text{arcsec } x, \sec^{-1} x$                                | Principal inverse secant.                  |
| $\text{arccot } x, \cot^{-1} x$                                | Principal inverse cotangent.               |
| $\sinh x$                                                      | Hyperbolic sine.                           |
| $\cosh x$                                                      | Hyperbolic cosine.                         |
| $\tanh x$                                                      | Hyperbolic tangent.                        |
| $\text{csch } x$                                               | Hyperbolic cosecant.                       |
| $\text{sech } x$                                               | Hyperbolic secant.                         |
| $\coth x$                                                      | Hyperbolic cotangent.                      |
| $\text{arcsinh } x, \text{arsinh } x,$<br>$\sinh^{-1} x$       | Principal inverse hyperbolic sine.         |
| $\text{arccosh } x, \text{arcosh } x,$<br>$\cosh^{-1} x$       | Principal inverse hyperbolic cosine.       |
| $\text{artanh } x, \text{artanh } x,$<br>$\tanh^{-1} x$        | Principal inverse hyperbolic tangent.      |
| $\text{arcsch } x, \text{arsch } x,$<br>$\text{csch}^{-1} x$   | Principal inverse hyperbolic cosecant.     |
| $\text{arcsech } x, \text{arsech } x,$<br>$\text{sech}^{-1} x$ | Principal inverse hyperbolic secant.       |
| $\text{arcoth } x, \text{arcoth } x,$<br>$\coth^{-1} x$        | Principal inverse hyperbolic cotangent.    |
| $\Gamma(z)$                                                    | Gamma function.                            |
| $B(x, y)$                                                      | Beta function.                             |
| $\zeta(s)$                                                     | Riemann zeta function.                     |

| Notation                                             | Meaning                                                                                 |
|------------------------------------------------------|-----------------------------------------------------------------------------------------|
| $D\mathbf{F}(\mathbf{a})$                            | Total derivative; Jacobian matrix in coordinates.                                       |
| $\text{Jac}(\mathbf{F})$                             | Jacobian determinant.                                                                   |
| $\text{Hess } f$                                     | Hessian matrix of $f$ .                                                                 |
| $\nabla f$                                           | Gradient of $f$ .                                                                       |
| $\nabla \cdot \mathbf{F}$                            | Divergence of $\mathbf{F}$ .                                                            |
| $\nabla \times \mathbf{F}$                           | Curl of $\mathbf{F}$ .                                                                  |
| $\Delta f$                                           | Laplacian of $f$ .                                                                      |
| $\mathbf{r}(t)$                                      | Parametrized curve.                                                                     |
| $ds$                                                 | Element of arc length.                                                                  |
| $\int_C \mathbf{F} \cdot d\mathbf{r}$                | Line integral along $C$ .                                                               |
| $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$          | Flux integral through $S$ .                                                             |
| $x_n \rightarrow x$                                  | Convergence of a sequence.                                                              |
| $a_n \nearrow L$                                     | Monotone increasing convergence to $L$ .                                                |
| $a_n \searrow L$                                     | Monotone decreasing convergence to $L$ .                                                |
| $\limsup a_n$                                        | Limit superior.                                                                         |
| $\liminf a_n$                                        | Limit inferior.                                                                         |
| $f_n \rightarrow f, f_n \xrightarrow{\text{ptw}} f$  | Pointwise convergence.                                                                  |
| $f_n \Rightarrow f, f_n \xrightarrow{\text{unif}} f$ | Uniform convergence.                                                                    |
| $f_n \xrightarrow{\text{a.e.}} f$                    | Convergence almost everywhere.                                                          |
| $C(\Omega), C^0(\Omega)$                             | Continuous functions on $\Omega$ .                                                      |
| $C^k(\Omega)$                                        | $k$ times continuously differentiable functions on $\Omega$ .                           |
| $C^\infty(\Omega)$                                   | Smooth functions on $\Omega$ .                                                          |
| $\mathcal{R}([a, b])$                                | Riemann-integrable functions on $[a, b]$ .                                              |
| $L^p(\Omega)$                                        | Lebesgue $L^p$ space on $\Omega$ .                                                      |
| $L^\infty(\Omega)$                                   | Lebesgue space of essentially bounded functions on $\Omega$ .                           |
| $\ f\ _\infty$                                       | Supremum norm $\sup_{x \in \Omega}  f(x) $ , when finite.                               |
| $\ f\ _{L^p(\Omega)}, \ f\ _p$                       | $L^p$ -norm $\left( \int_\Omega  f(x) ^p \, dx \right)^{1/p}$ for $1 \leq p < \infty$ . |
| $\ f\ _{L^\infty(\Omega)}$                           | $L^\infty$ -norm $\text{ess sup}_{x \in \Omega}  f(x) $ .                               |
| $B_r(x)$                                             | Open ball of radius $r$ centered at $x$ .                                               |
| $\overline{B}_r(x)$                                  | Closed ball of radius $r$ centered at $x$ .                                             |
| $z = x + iy$                                         | Complex number with real part $x$ and imaginary part $y$ .                              |
| $\text{Re } z$                                       | Real part of $z$ .                                                                      |
| $\text{Im } z$                                       | Imaginary part of $z$ ; distinct from lowercase im for image.                           |
| $\bar{z}$                                            | Complex conjugate.                                                                      |
| $ z $                                                | Complex modulus.                                                                        |
| $\arg z$                                             | Argument of $z$ .                                                                       |
| $\text{Res}(f; a), \text{Res}_{z=a} f(z)$            | Residue of $f$ at $a$ .                                                                 |

## 5 Linear Algebra

| Notation                                                           | Meaning                                                                                              |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| $\mathbf{A} = (a_{ij})$                                            | Matrix with $(i, j)$ -entry $a_{ij}$ .                                                               |
| $\mathbf{A}^\top$                                                  | Transpose of $\mathbf{A}$ .                                                                          |
| $\mathbf{A}^H$                                                     | Conjugate transpose of $\mathbf{A}$ .                                                                |
| $\mathbf{A}^{-1}$                                                  | Inverse of $\mathbf{A}$ .                                                                            |
| $\mathbf{I}_n, \mathbf{I}$                                         | Identity matrix; the dimension subscript is preferred when readily available and useful.             |
| $\mathbf{0}$                                                       | Zero vector or zero matrix, with dimension understood.                                               |
| $\mathbf{D}$                                                       | Diagonal matrix when a local name is required.                                                       |
| $\mathbf{P}$                                                       | Projection matrix when a local name is required.                                                     |
| $\text{diag}(d_1, \dots, d_n)$                                     | Diagonal matrix with entries $d_1, \dots, d_n$ .                                                     |
| $\mathbf{1}$                                                       | All-ones vector.                                                                                     |
| $e^{\mathbf{A}}, \exp(\mathbf{A})$                                 | Matrix exponential; $e^{t\mathbf{A}}$ is the standard time-dependent form.                           |
| $\det(\mathbf{A}),  \mathbf{A} $                                   | Determinant.                                                                                         |
| $\text{rk}(\mathbf{A}), \text{rank}(\mathbf{A})$                   | Rank.                                                                                                |
| $\text{tr}(\mathbf{A})$                                            | Trace.                                                                                               |
| $\text{adj}(\mathbf{A})$                                           | Adjugate matrix.                                                                                     |
| $\text{Col}(\mathbf{A})$                                           | Column space.                                                                                        |
| $\text{Row}(\mathbf{A})$                                           | Row space.                                                                                           |
| $\text{Null}(\mathbf{A}), \ker \mathbf{A}$                         | Null space.                                                                                          |
| $\ker T$                                                           | Kernel of a linear map $T$ .                                                                         |
| $\text{im } T$                                                     | Image of a linear map $T$ .                                                                          |
| $\mathbf{e}_i$                                                     | $i$ th standard basis vector.                                                                        |
| $\dim V$                                                           | Dimension of $V$ .                                                                                   |
| $\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$                   | Span of the listed vectors.                                                                          |
| $U \oplus V$                                                       | Direct sum.                                                                                          |
| $\langle \mathbf{u}, \mathbf{v} \rangle, (\mathbf{u}, \mathbf{v})$ | Inner product; over complex spaces, conjugate-linear in the first argument and linear in the second. |
| $\mathbf{u} \cdot \mathbf{v}, \mathbf{u}^\top \mathbf{v}$          | Standard Euclidean inner product on $\mathbb{R}^n$ .                                                 |
| $\mathbf{u}^H \mathbf{v}$                                          | Standard Hermitian inner product on $\mathbb{C}^n$ .                                                 |
| $\ \mathbf{v}\ $                                                   | Norm of $\mathbf{v}$ .                                                                               |
| $\mathbf{u} \perp \mathbf{v}$                                      | Orthogonality.                                                                                       |
| $U^\perp$                                                          | Orthogonal complement of the subspace $U$ .                                                          |
| $\text{proj}_U \mathbf{x}$                                         | Orthogonal projection of $\mathbf{x}$ onto the subspace $U$ .                                        |
| $\text{proj}_{\mathbf{v}} \mathbf{x}$                              | Orthogonal projection of $\mathbf{x}$ onto $\text{Span}(\mathbf{v})$ .                               |
| $\mathbf{u} \times \mathbf{v}$                                     | Cross product in $\mathbb{R}^3$ .                                                                    |
| $p_{\mathbf{A}}(\lambda), \chi_{\mathbf{A}}(\lambda)$              | Characteristic polynomial of $\mathbf{A}$ .                                                          |
| $m_{\mathbf{A}}(\lambda), \mu_{\mathbf{A}}(\lambda)$               | Minimal polynomial of $\mathbf{A}$ .                                                                 |
| $\lambda, \lambda_i$                                               | Eigenvalue of $\mathbf{A}$ .                                                                         |
| $\mathbf{v}, \mathbf{v}_i$                                         | Eigenvector of $\mathbf{A}$ .                                                                        |
| $E_\lambda(\mathbf{A}), V_\lambda(\mathbf{A})$                     | Eigenspace of $\mathbf{A}$ associated with $\lambda$ .                                               |
| $\sigma(\mathbf{A}), \text{spec}(\mathbf{A})$                      | Spectrum of $\mathbf{A}$ .                                                                           |
| $\rho(\mathbf{A})$                                                 | Spectral radius.                                                                                     |
| $\sigma_i(\mathbf{A})$                                             | $i$ th singular value of $\mathbf{A}$ .                                                              |



## 6 Differential Equations

The Fourier transform uses the normalization

$$\widehat{f}(\boldsymbol{\xi}) := \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} d\mathbf{x}, \quad f(\mathbf{x}) = \int_{\mathbb{R}^n} \widehat{f}(\boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} d\boldsymbol{\xi},$$

whenever the Fourier inversion formula applies.

| Notation                                                                                                           | Meaning                                                                                                |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $y'$                                                                                                               | First derivative with respect to the understood independent variable.                                  |
| $y''$                                                                                                              | Second derivative.                                                                                     |
| $\dot{x}$                                                                                                          | First derivative of $x$ with respect to time.                                                          |
| $\ddot{x}$                                                                                                         | Second derivative of $x$ with respect to time.                                                         |
| $y_h, y_c$                                                                                                         | Homogeneous or complementary part of a solution.                                                       |
| $y_p$                                                                                                              | Particular solution.                                                                                   |
| $c_1, c_2, \dots$                                                                                                  | Arbitrary constants in a general solution.                                                             |
| $W(y_1, \dots, y_n)(t),$<br>$W[y_1, \dots, y_n](t)$                                                                | Wronskian of $y_1, \dots, y_n$ .                                                                       |
| $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{g}(t),$<br>$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$ | Linear first-order system in state-vector form.                                                        |
| $\Phi(t)$                                                                                                          | Fundamental matrix of a linear system.                                                                 |
| $\mathcal{L}\{f(t)\}(s),$<br>$\mathcal{L}[f](s)$                                                                   | Laplace transform of $f$ .                                                                             |
| $\mathcal{L}^{-1}\{F(s)\}(t),$<br>$\mathcal{L}^{-1}[F](t)$                                                         | Inverse Laplace transform of $F$ .                                                                     |
| $(f * g)(t)$                                                                                                       | One-sided convolution, $(f * g)(t) := \int_0^t f(\tau)g(t - \tau) d\tau$ .                             |
| $H(t - a), u(t - a)$                                                                                               | Heaviside or unit-step function shifted to $t = a$ ; its value at the jump is specified when relevant. |
| $G(x, \xi)$                                                                                                        | Green's function; the variables depend on context.                                                     |
| $J_\nu(x)$                                                                                                         | Bessel function of the first kind of order $\nu$ .                                                     |
| $Y_\nu(x), N_\nu(x)$                                                                                               | Bessel function of the second kind (Neumann function) of order $\nu$ .                                 |
| $\widehat{f}(\boldsymbol{\xi}), \mathcal{F}f(\boldsymbol{\xi})$                                                    | Fourier transform under the normalization above.                                                       |

## 7 Combinatorics and Discrete Mathematics

| Notation                             | Meaning                                                                       |
|--------------------------------------|-------------------------------------------------------------------------------|
| $\binom{n}{r}, {}_nC_r$              | Number of $r$ -element subsets of an $n$ -element set.                        |
| $\left(\binom{n}{r}\right), {}_nH_r$ | $r$ -combinations with repetition from $n$ types; equals $\binom{n+r-1}{r}$ . |
| $n!$                                 | Factorial.                                                                    |
| $n^{\overline{k}}, (n)_k$            | Falling factorial.                                                            |
| $n^{\overline{k}}, n^{(k)}$          | Rising factorial.                                                             |
| $O(g(n))$                            | Big-O upper asymptotic bound.                                                 |
| $o(g(n))$                            | Little-o strict asymptotic bound.                                             |
| $\Theta(g(n))$                       | Two-sided asymptotic order.                                                   |
| $\Omega(g(n))$                       | Asymptotic lower bound.                                                       |
| $G = (V, E)$                         | Graph with vertex set $V$ and edge set $E$ .                                  |

| Notation          | Meaning                         |
|-------------------|---------------------------------|
| $V(G)$            | Vertex set of $G$ .             |
| $E(G)$            | Edge set of $G$ .               |
| $N(v), \Gamma(v)$ | Neighborhood of vertex $v$ .    |
| $\deg(v)$         | Degree of vertex $v$ .          |
| $K_n$             | Complete graph on $n$ vertices. |
| $P_n$             | Path graph on $n$ vertices.     |
| $C_n$             | Cycle graph on $n$ vertices.    |

## 8 Probability and Statistics

Probability and expectation are written in blackboard bold by default.

| Notation                                                          | Meaning                                                       |
|-------------------------------------------------------------------|---------------------------------------------------------------|
| $\mathbb{P}(A), P(A)$                                             | Probability of event $A$ .                                    |
| $\mathbb{P}(A \mid B), P(A \mid B)$                               | Conditional probability of $A$ given $B$ .                    |
| $X \perp\!\!\!\perp Y$                                            | $X$ and $Y$ are independent.                                  |
| $X \perp\!\!\!\perp Y \mid Z$                                     | $X$ and $Y$ are conditionally independent given $Z$ .         |
| $\mathbb{E}[X], E[X]$                                             | Expectation of random variable $X$ .                          |
| $\mathbb{E}[X \mid Y], E[X \mid Y]$                               | Conditional expectation of $X$ given $Y$ .                    |
| $\text{Var}(X)$                                                   | Variance of $X$ .                                             |
| $\text{Cov}(X, Y)$                                                | Covariance of $X$ and $Y$ .                                   |
| $X \sim \text{Bin}(n, p)$                                         | Binomial distribution.                                        |
| $X \sim \text{Pois}(\lambda)$                                     | Poisson distribution.                                         |
| $X \sim \mathcal{N}(\mu, \sigma^2),$<br>$X \sim N(\mu, \sigma^2)$ | Normal distribution with mean $\mu$ and variance $\sigma^2$ . |
| $X_n \xrightarrow{\text{a.s.}} X$                                 | Almost-sure convergence.                                      |
| $X_n \xrightarrow{\mathbb{P}} X$                                  | Convergence in probability.                                   |
| $X_n \xrightarrow{L^p} X$                                         | Convergence in $L^p$ .                                        |
| $X_n \xrightarrow{d} X, X_n \Rightarrow X$                        | Convergence in distribution.                                  |

## 9 Geometry

In triangle notation,  $A, B, C$  denote vertices and  $a, b, c$  the opposite side lengths.

| Notation                    | Meaning                                |
|-----------------------------|----------------------------------------|
| $\triangle ABC$             | Triangle with vertices $A, B, C$ .     |
| $\angle ABC$                | Angle with vertex $B$ .                |
| $\overline{AB}$             | Segment joining $A$ and $B$ .          |
| $AB$                        | Length of $\overline{AB}$ .            |
| $[ABC]$                     | Area of $\triangle ABC$ .              |
| $\text{length}(C), \ell(C)$ | Length of $C$ .                        |
| $\text{area}(S)$            | Area of $S$ .                          |
| $\text{vol}(S)$             | Volume of $S$ .                        |
| $\text{dist}(P, S)$         | Distance from $P$ to $S$ .             |
| $\text{conv}(S)$            | Convex hull of $S$ .                   |
| $\mathbb{R}^n$              | $n$ -dimensional real Euclidean space. |

| Notation       | Meaning                            |
|----------------|------------------------------------|
| $\mathbb{H}^n$ | $n$ -dimensional hyperbolic space. |
| $\parallel$    | Parallel.                          |
| $\perp$        | Perpendicular or orthogonal.       |

## 10 Number Theory

| Notation                          | Meaning                                         |
|-----------------------------------|-------------------------------------------------|
| $a \mid b$                        | $a$ divides $b$ .                               |
| $a \nmid b$                       | $a$ does not divide $b$ .                       |
| $\gcd(a, b)$                      | Greatest common divisor, taken nonnegative.     |
| $\text{lcm}(a, b)$                | Least common multiple.                          |
| $a \equiv b \pmod{n}$             | $a$ is congruent to $b$ modulo $n$ .            |
| $[a]_n$                           | Residue class of $a$ modulo $n$ .               |
| $\mathbb{Z}/n\mathbb{Z}$          | Ring of residue classes modulo $n$ .            |
| $(\mathbb{Z}/n\mathbb{Z})^\times$ | Multiplicative group of units modulo $n$ .      |
| $a^{-1} \pmod{n}$                 | Multiplicative inverse of $a$ modulo $n$ .      |
| $\left(\frac{a}{p}\right)$        | Legendre symbol.                                |
| $\varphi(n)$                      | Euler totient function.                         |
| $\text{ord}_n(a)$                 | Multiplicative order of $a$ modulo $n$ .        |
| $\nu_p(n), \nu_p(n)$              | $p$ -adic valuation of $n$ .                    |
| $\tau(n)$                         | Number of positive divisors of $n$ .            |
| $\sigma(n)$                       | Sum of positive divisors of $n$ .               |
| $\mu(n)$                          | Möbius function.                                |
| $\text{rad}(n)$                   | Product of the distinct prime divisors of $n$ . |
| $\pi(x)$                          | Number of primes not exceeding $x$ .            |
| $\mathbb{F}_q, \text{GF}(q)$      | Finite field with $q$ elements.                 |

## 11 Algebra

$G, H$  usually denote groups,  $R, S$  rings, and  $F, K, L$  fields.

| Notation              | Meaning                           |
|-----------------------|-----------------------------------|
| $G$                   | Group.                            |
| $e, 1_G$              | Identity element of a group.      |
| $ G $                 | Order of a finite group.          |
| $\langle S \rangle$   | Subgroup generated by $S$ .       |
| $H \leq G$            | $H$ is a subgroup of $G$ .        |
| $H \trianglelefteq G$ | $H$ is a normal subgroup of $G$ . |
| $[G : H]$             | Index of $H$ in $G$ .             |
| $G/H$                 | Quotient group.                   |
| $\ker \varphi$        | Kernel of $\varphi$ .             |
| $\text{im } \varphi$  | Image of $\varphi$ .              |
| $\text{Aut}(G)$       | Automorphism group of $G$ .       |
| $Z(G)$                | Center of $G$ .                   |
| $C_G(x)$              | Centralizer of $x$ in $G$ .       |

| Notation                                             | Meaning                                         |
|------------------------------------------------------|-------------------------------------------------|
| $\mathrm{GL}_n(F)$                                   | General linear group over $F$ .                 |
| $\mathrm{SL}_n(F)$                                   | Special linear group over $F$ .                 |
| $S_n, \mathfrak{S}_n$                                | Symmetric group on $n$ symbols.                 |
| $A_n, \mathfrak{A}_n$                                | Alternating group on $n$ symbols.               |
| $R$                                                  | Ring.                                           |
| $I \trianglelefteq R$                                | $I$ is an ideal of $R$ .                        |
| $(a_1, \dots, a_n), \langle a_1, \dots, a_n \rangle$ | Ideal generated by $a_1, \dots, a_n$ .          |
| $R/I$                                                | Quotient ring.                                  |
| $F$                                                  | Base field in an algebraic context.             |
| $K/F$                                                | Field extension of $F$ by $K$ .                 |
| $F[x]$                                               | Polynomial ring in $x$ over $F$ .               |
| $\deg f$                                             | Degree of a polynomial $f$ .                    |
| $\mathrm{char} F$                                    | Characteristic of $F$ .                         |
| $\mathbb{H}$                                         | Quaternion division algebra over $\mathbb{R}$ . |
| $\mathrm{Gal}(L/K)$                                  | Galois group of $L/K$ .                         |
| $\mathrm{trdeg}_K L$                                 | Transcendence degree of $L$ over $K$ .          |

## 12 Topology

| Notation                      | Meaning                                             |
|-------------------------------|-----------------------------------------------------|
| $(X, \mathcal{T})$            | Topological space $X$ with topology $\mathcal{T}$ . |
| $A^\circ, \mathrm{int} A$     | Interior of $A$ .                                   |
| $\overline{A}, \mathrm{cl} A$ | Closure of $A$ .                                    |
| $\partial A$                  | Boundary of $A$ .                                   |
| $\pi_1(X, x_0)$               | Fundamental group of $X$ based at $x_0$ .           |
| $\chi(X)$                     | Euler characteristic of $X$ .                       |
| $X \cong Y$                   | Homeomorphic spaces.                                |
| $X \simeq Y$                  | Homotopy equivalent spaces.                         |

## 13 Miscellaneous Notation

| Notation            | Meaning                                                     |
|---------------------|-------------------------------------------------------------|
| $\lfloor x \rfloor$ | Floor of $x$ .                                              |
| $\lceil x \rceil$   | Ceiling of $x$ .                                            |
| $\{x\}$             | Fractional part $x - \lfloor x \rfloor$ , when unambiguous. |
| $\mathrm{sgn}(x)$   | Sign of a real number or permutation, according to context. |
| $\Delta q$          | Finite change in a quantity $q$ .                           |
| $\epsilon_{ijk}$    | Levi-Civita symbol.                                         |